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Project Title:	Control of Bacterial Biofilm Formation in Space Exploration		
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Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
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Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
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Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
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Flight Program:			
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Task Description:	Bacteria in nature and disease often choose a mode of growth known as a biofilm, which consists of numerous aggregating microorganisms (microcolonies) attached to a surface. Biofilm formation is a leading cause of bacterial pneumonia which can be acquired by astronauts during a long-term space mission and by patients with compromised immune systems such as those with cystic fibrosis. A capsular material called mucoid exopolysaccharide, or alginate is important in the formation of biofilms because it provides a key structural component for in vivo biofilms. The objective of this proposal is to study the genetic regulation of alginate in biofilm formation. The model organism is Pseudomonas aeruginosa, a ubiquitous biofilm-forming bacterium capable of displaying two distinctive phenotypes: a nonmucoid or mucoid colony morphology. The mucoid form, which is due to the overproduction of alginate, is a clinical indicator for a poor prognosis. While the two morphologies can be made inter-changeable under the laboratory conditions, little is known about the molecular mechanisms governing this process. Using genetics and functional genomics approaches, we will search for the biofilm “switches” on the genomes. The identified genes may become targets for the development of anti-biofilm therapeutics. The specific aims of this project are i) to identify and characterize the genes in the alginate regulation, ii) to elucidate the underlying mechanisms, and iii) determine what/how environmental signals modulate the expression of these regulators. Our long-term goal is to target bacterial own genes for the control of the space biofilm formation. The objective of the proposed research is to identify and characterize the biofilm-controlling genetic elements from the genomes of P. aeruginosa. It is our expectation that the approaches used here will lead us to identify the genes that control the formation of the mucoid biofilms. We will also provide new information regarding how these genes control the phenotypic switch, and what/how environmental signals may stimulate or inhibit the expression of these genes. These results will be significant, because they will provide genetic targets for anti-biofilm therapeutics, and aid in the future development of the gene-based space medicine. In addition, it is expected that the results will advance the emerging field of microbial biofilms.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The major Earth benefits of this proposal are to provide the basis for the future development of the new genetically defined drug with improved specificity and function and very little side effect.
Task Progress:	Identification of three new regulators for alginate biosynthesis. We have continued the global transposon screen to search for the novel regulators for mucoid biofilms in P. aeruginosa. Instead, screening the wild-type nonmucoid to mucoid conversion, we took a reverse approach by screening for conversion from stable, mucoid mutant to nonmucoid reversion due to the transposon mutagenesis. We identified three new genes that serve as positive regulators for alginate biosynthesis. It is known that alginate production is regulated by the ECF sigma factor, AlgU/T, and the cognate anti-sigma factor, MucA. Many clinical mucoid isolates carry the loss-of-function mutations in mucA. These mutations, including the most common mucA22 allele, cause C-terminal truncations in MucA, indicating an inability to regulate the AlgU activity by MucA is associated with conversion to the mucoid phenotype. We found that a mutation in a stable mucoid strain derived from the parental strain PAO1, designated PAO581, that does not contain the mucA22 allele, was due to a single base deletion in mucA (delta T180), generating another type of C-terminal truncation. A global mariner transposon screen in PAO581 for non-mucoid isolates led to the identification of three regulators of alginate production, clpP (PA1801), clpX (PA1802), and a clpP paralog (PA3326, designated clpP2). The PAO581 null mutants of clpP, clpX, and clpP2 showed a decrease in AlgU transcriptional activity and an accumulation of HA-tagged, N-terminal MucA protein with an apparent molecular mass of 15kDa. The clpP and clpX mutants of a CF mucoid isolate revert to the non-mucoid phenotype. The ClpXP and ClpP2 proteins appear to be part of a proteolytic network degrading the cytoplasmic portion of truncated MucA proteins to release the sequestered AlgU, which drives alginate biosynthesis. This work has been published in Microbiology (Qiu, D., V. M. Eisinger, N. E. Head, G. B. Pier, and H. D. Yu. 2008. ClpXP proteases positively regulate alginate over-expression and mucoid conversion in Pseudomonas aeruginosa. Microbiol. 154: 2119-2130). Development of a set of versatile and controllable expression vectors. We have reported the construction of a series of Escherichia-Pseudomonas broad-host-range expression vectors utilizing the PBAD promoter and the araC regulator for routine cloning, conditional expression, and analysis of tightly controlled and/or toxic genes in pseudomonads. This work has been published in Applied and Environmental Microbiology (Qiu, D., F. H. Damron, T. Mima, H. P. Schweizer, and H. D. Yu. 2008. PBAD-based shuttle vectors for functional analysis of toxic and highly-regulated genes in Pseudomonas and Burkholderia spp. and other bacteria. Appl. Environ. Microbiol.74: 7422-7426). Identification of a new mechanism regulating alginate overproduction in P. aeruginosa. Mucoidy or overproduction of the exopolysaccharide known as alginate in Pseudomonas aeruginosa is a poor prognosticator for lung infections in cystic fibrosis. Mutation of the anti-s factor MucA is a well-accepted mechanism for mucoid conversion. However, certain clinical mucoid strains of P. aeruginosa have a wild-type (wt) mucA. Here, we describe a loss-of-function mutation in kinB that causes overproduction of alginate in the wt mucA strain PAO1. KinB is the cognate histidine kinase for the transcriptional activator AlgB. Increased alginate production due to inactivation of kinB was correlated with the high expression at the alginate-related promoters PalgU and PalgD. Deletion of alternative s factor RpoN (?54) or the response regulator AlgB in kinB mutants decreases alginate production to wt nonmucoid levels. Mucoidy was restored in the kinB algB double mutant by expression of wt AlgB or phosphorylation defective AlgB.D59N, indicating that phosphorylation of AlgB was not required for alginate overproduction when kinB was inactivated. The inactivation of the DegS-like protease algW in the kinB mutant caused loss of alginate production and an accumulation of the hemagglutinin (HA)-tagged MucA. Furthermore, we observed the kinB mutation increased the rate of HA-MucA degradation. Our results also indicate AlgW-mediated MucA degradation requires algB and rpoN in the kinB mutant. Collectively, these studies indicate KinB is a negative regulator of alginate production in wt mucA strain PAO1. This paper has been accepted for publication by Journal of Bacteriology (Damron, F. H., D. Qiu, and H. D. Yu. Pseudomonas aeruginosa sensor kinase KinB negatively controls alginate production through AlgW-dependent MucA proteolysis. J. Bacteriol. Accepted). Identification of a stress-responsive novel lipoprotein in mucoid, alginate overproducing P. aeruginosa. Chronic lung infection with P. aeruginosa and excessive neutrophils-associated inflammation are the major causes of morbidity and mortality in patients with cystic fibrosis (CF). Overproduction of an exopolysaccharide known as alginate leads to the

	formation of mucoid biofilms that are resistant to antibiotics and host defenses. Alginate overproduction or mucoidy is controlled by a stress-related ECF sigma factor AlgU/T. Mutation in the anti-sigma factor MucA is a known mechanism for conversion to mucoidy. Recently, we identified inactivation of a kinase (KinB) in PAO1 caused overproduction of alginate. Here we report the identification of lipotoxin F (LptF, PA3692), an outer membrane protein F paralog whose expression increased in PAO1 kinB mutant (PAO1kinB::aacC1). The lipotoxin family was previously characterized to induce inflammation in lung epithelia. Inactivation of lptF had no effect on mucoidy. Deletion of lptF caused a differential susceptibility to phagocyte-generated oxidants. The lptF, algU and algW mutants were more sensitive to hypochlorite than parent PAO1. However, the lptF mutant displayed increased resistance to hydrogen peroxide, indicating LptF may mediate resistance to bactericidal activity of neutrophils. Expression of LptF was AlgU-dependent and observed in CF isolates. LptF also mediated adhesion to human lung epithelial cells (A549). The lptF promoter activity increased upon exposure to ultraviolet irradiation (UV-B). Collectively, LptF, a stress response marker, may be required for P. aeruginosa survival in harsh environments including lung colonization in CF. The paper has been submitted and is currently in revision (Damron, F. H., J. Napper, M. A. Teter, and H. D. Yu. Lipotoxin F in Pseudomonas aeruginosa is an AlgU-dependent and alginate independent stress response marker that mediates resistance to oxidative stress and adhesion to A549 human lung epithelia. Microbiol. In revision). Submission of the second patent application for developing novel methods for producing bacterial alginates. Currently, alginate is made from seaweeds. The large-scale sea farming of seaweeds poses a direct threat to the environment. In addition, the level of polymerization, monosaccharide polydispersity of seaweed alginates cannot be controlled. Bioprocessing and separation of certain types of seaweed alginates are expensive. Both strains of high alginate producers that are patent-pending are being utilized by Progenesis Technologies, LLC as prototypes for commercialization of bacterial alginates. Progenesis Technologies, LLC is a startup biotech company co-founded by Dr. Niles and Dr. Yu from Marshall University and has acquired the exclusive license from Marshall University Research Corporation to utilize this technology for the production of alginates. Since its founding in 2006, Progenesis has raised enough capitals from private investors. It has been in operation since Feb. 2008, and located in a biotech business incubator managed by the Chemical Alliance Zone (CAZ) in South Charleston, West Virginia.

Bibliography Type:	Description: (Last Updated: 04/09/2020)
Articles in Peer-reviewed Journals	Qiu D, Eisinger VM, Head NE, Pier GB, Yu HD. "ClpXP proteases positively regulate alginate overexpression and mucoid conversion in Pseudomonas aeruginosa." Microbiology. 2008 Jul;154(Pt 7):2119-30. PMID: 18599839 , Jul-2008
Articles in Peer-reviewed Journals	Qiu D, Damron FH, Mima T, Schweizer HP, Yu HD. "PBAD-based shuttle vectors for functional analysis of toxic and highly regulated genes in Pseudomonas and Burkholderia spp. and other bacteria." Appl Environ Microbiol. 2008 Dec;74(23):7422-6. PMID: 18849445 , Dec-2008
Articles in Peer-reviewed Journals	Damron FH, Qiu D, Yu HD. "Pseudomonas aeruginosa sensor kinase KinB negatively controls alginate production through AlgW-dependent MucA proteolysis." Journal of Bacteriology, accepted status, as of January 2009. , Jan-2009
Articles in Peer-reviewed Journals	Damron FH, Yu HD. "Pseudomonas aeruginosa MucD regulates the alginate pathway through activation of MucA degradation via MucP proteolytic activity. " J Bacteriol. 2011 Jan;193(1):286-91. Epub 2010 Oct 29. https://doi.org/10.1128/JB.01132-10 ; PubMed PMID: 21036998 ; PubMed Central PMCID: PMC3019965 , Jan-2011
Articles in Peer-reviewed Journals	Damron FH, Napper J, Teter MA, Yu HD. "Lipotoxin F of Pseudomonas aeruginosa is an AlgU-dependent and alginate-independent outer membrane protein involved in resistance to oxidative stress and adhesion to A549 human lung epithelia." Microbiology. 2009 Apr;155(Pt 4):1028-1038. https://doi.org/10.1099/mic.0.025833-0 ; PubMed PMID: 19332805 ; PubMed Central PMCID: PMC2895231 , Apr-2009
Patents	61/048,858. Pending, April 2008. Apr-2008 Yu HD, Qiu D. "Methods of Producing Bacterial Alginates."